Waste Management System Test Cases and Results

Software Quality Assurance and Testing | Coventry University

This document presents each identified automated test case in a four-column format. Test scenarios are taken from the test source inventory. Expected outcomes are plain-language descriptions based on the test names. Detailed assertions remain in the corresponding source files.

Recorded execution summary

Test level	Previously reported result
Unit / API	307 of 307 passed
Integration	114 of 114 passed
End-to-End	20 of 20 passed

Unit / API test cases

Test ID	**Test scenario**	**Expected outcome**
TC-ADM-APP-01	approves a pending request successfully	The system successfully approves a pending request successfully.
TC-ADM-APP-02	returns 401 for an unauthenticated user	The system returns 401 for an unauthenticated user.
TC-ADM-APP-03	returns 403 when a Resident attempts approval	The system returns 403 when a Resident attempts approval.
TC-ADM-APP-04	returns 400 when request ID is empty	The system returns 400 when request ID is empty.
TC-ADM-APP-05	returns 400 for whitespace-only request ID	The system returns 400 for whitespace-only request ID.
TC-ADM-APP-06	returns 404 when the request does not exist	The system returns 404 when the request does not exist.

Test ID	Test scenario	Expected outcome
TC-ADM-APP-07	returns 409 for an already approved request	The system returns 409 for an already approved request.
TC-ADM-APP-08	rejects approval of a cancelled request	The system rejects approval of a cancelled request, without allowing the prohibited action.
TC-ADM-APP-09	rejects approval of a rejected request	The system rejects approval of a rejected request, without allowing the prohibited action.
TC-ADM-APP-10	rejects approval of an assigned request	The system rejects approval of an assigned request, without allowing the prohibited action.
TC-ADM-APP-11	rejects approval of an in-progress request	The system rejects approval of an in-progress request, without allowing the prohibited action.
TC-ADM-APP-12	rejects approval of a completed request	The system rejects approval of a completed request, without allowing the prohibited action.
TC-ADM-APP-13	conditionally updates status and approval timestamp	The system successfully conditionally updates status and approval timestamp.

Test ID	Test scenario	Expected outcome
TC-ADM-APP-14	creates an approval activity record	The system successfully creates an approval activity record.
TC-ADM-APP-15	returns 500 when the database lookup fails	The system returns 500 when the database lookup fails.
TC-ADM-APP-16	returns 500 when the transaction fails	The system returns 500 when the transaction fails.
TC-ADM-APP-17	returns 500 when activity creation fails	The system returns 500 when activity creation fails.
TC-ADM-APP-18	does not create an activity when the update fails	The system does not create an activity when the update fails.
TC-ADM-APP-19	prevents duplicate approval when status changes concurrently	The system prevents duplicate approval when status changes concurrently, without allowing the prohibited action.
TC-ADM-ASG-01	assigns an active collector successfully	The system successfully assigns an active collector successfully.
TC-ADM-ASG-02	rejects unauthenticated users	The system rejects unauthenticated users, without allowing the prohibited action.
TC-ADM-ASG-03	rejects non-admin users	The system rejects non-admin users, without allowing the prohibited action.

Test ID	Test scenario	Expected outcome
TC-ADM-ASG-04	rejects an empty request ID	The system rejects an empty request ID, without allowing the prohibited action.
TC-ADM-ASG-05	rejects a whitespace-only request ID	The system rejects a whitespace-only request ID, without allowing the prohibited action.
TC-ADM-ASG-06	rejects malformed JSON	The system rejects malformed JSON, without allowing the prohibited action.
TC-ADM-ASG-07	rejects a missing collector ID	The system rejects a missing collector ID, without allowing the prohibited action.
TC-ADM-ASG-08	rejects a whitespace-only collector ID	The system rejects a whitespace-only collector ID, without allowing the prohibited action.
TC-ADM-ASG-09	returns 404 for a nonexistent request	The system returns 404 for a nonexistent request.
TC-ADM-ASG-13	rejects a nonexistent collector	The system rejects a nonexistent collector, without allowing the prohibited action.
TC-ADM-ASG-14	rejects an inactive collector	The system rejects an inactive collector, without allowing the prohibited action.
TC-ADM-ASG-15	queries only active users with the collector role	The system successfully queries only active users with the collector role.
TC-ADM-ASG-16	atomically updates collector, status and assignment timestamp	The system successfully atomically updates collector, status and assignment timestamp.
TC-ADM-ASG-17	creates an assignment activity linked to the admin	The system successfully creates an assignment activity linked to the admin.
TC-ADM-ASG-18	returns the collector	The system returns the collector.
TC-ADM-ASG-19	returns 500 when the request lookup fails	The system returns 500 when the request lookup fails.

Test ID	Test scenario	Expected outcome
TC-ADM-ASG-20	returns 500 when the transaction fails	The system returns 500 when the transaction fails.
TC-ADM-ASG-21	rejects a null JSON body with HTTP 400	The system rejects a null JSON body with HTTP 400, without allowing the prohibited action.
TC-ADM-ASG-22	rejects a numeric collector ID with HTTP 400	The system rejects a numeric collector ID with HTTP 400, without allowing the prohibited action.
TC-ADM-ASG-23	prevents assignment when another operation has already changed the status	The system prevents assignment when another operation has already changed the status, without allowing the prohibited action.
TC-ADM-COL-01	creates a valid collector	The system successfully creates a valid collector.
TC-ADM-COL-02	rejects unauthenticated requests	The system rejects unauthenticated requests, without allowing the prohibited action.
TC-ADM-COL-03	rejects resident users	The system rejects resident users, without allowing the prohibited action.
TC-ADM-COL-04	rejects collector users	The system rejects collector users, without allowing the prohibited action.
TC-ADM-COL-05	rejects missing first name	The system rejects missing first name, without allowing the prohibited action.
TC-ADM-COL-06	rejects missing last name	The system rejects missing last name, without allowing the prohibited action.
TC-ADM-COL-07	rejects missing email	The system rejects missing email, without allowing the prohibited action.
TC-ADM-COL-08	rejects missing zone	The system rejects missing zone, without allowing the prohibited action.

Test ID	Test scenario	Expected outcome
TC-ADM-COL-09	rejects whitespace-only first name	The system rejects whitespace-only first name, without allowing the prohibited action.
TC-ADM-COL-10	rejects whitespace-only zone	The system rejects whitespace-only zone, without allowing the prohibited action.
TC-ADM-COL-11	rejects invalid email format	The system rejects invalid email format, without allowing the prohibited action.
TC-ADM-COL-12	rejects non-string first name	The system rejects non-string first name, without allowing the prohibited action.
TC-ADM-COL-13	rejects duplicate email	The system rejects duplicate email, without allowing the prohibited action.
TC-ADM-COL-14	normalises email before duplicate lookup	The system successfully normalises email before duplicate lookup.
TC-ADM-COL-15	generates a temporary password	The system successfully generates a temporary password.
TC-ADM-COL-16	hashes the temporary password	The system successfully hashes the temporary password.
TC-ADM-COL-17	requires password change	The system successfully requires password change.
TC-ADM-COL-18	assigns the COLLECTOR role	The system successfully assigns the COLLECTOR role.
TC-ADM-COL-19	creates collector profile with correct zone	The system successfully creates collector profile with correct zone.
TC-ADM-COL-20	uses a transaction for account and profile creation	The system successfully uses a transaction for account and profile creation.
TC-ADM-COL-21	handles password hashing failure	The system handles password hashing failure without producing an uncontrolled failure.

Test ID	Test scenario	Expected outcome
TC-ADM-COL-22	handles user insertion failure	The system handles user insertion failure without producing an uncontrolled failure.
TC-ADM-COL-23	handles collector profile insertion failure	The system handles collector profile insertion failure without producing an uncontrolled failure.
TC-ADM-COL-24	rejects malformed JSON	The system rejects malformed JSON, without allowing the prohibited action.
TC-ADM-COL-25	returns the expected success response	The system returns the expected success response.
TC-ADM-COL-26	does not log temporary passwords	The system does not log temporary passwords.
TC-ADM-USR-01	creates a valid resident	The system successfully creates a valid resident.
TC-ADM-USR-02	rejects unauthenticated requests	The system rejects unauthenticated requests, without allowing the prohibited action.
TC-ADM-USR-03	rejects resident users	The system rejects resident users, without allowing the prohibited action.
TC-ADM-USR-04	rejects collector users	The system rejects collector users, without allowing the prohibited action.
TC-ADM-USR-05	rejects missing first name	The system rejects missing first name, without allowing the prohibited action.
TC-ADM-USR-06	rejects missing last name	The system rejects missing last name, without allowing the prohibited action.
TC-ADM-USR-07	rejects missing email	The system rejects missing email, without allowing the prohibited action.
TC-ADM-USR-08	rejects whitespace-only names	The system rejects whitespace-only names, without allowing the prohibited action.

Test ID	Test scenario	Expected outcome
TC-ADM-USR-09	rejects invalid email format	The system rejects invalid email format, without allowing the prohibited action.
TC-ADM-USR-10	rejects duplicate email	The system rejects duplicate email, without allowing the prohibited action.
TC-ADM-USR-11	normalises email before duplicate lookup	The system successfully normalises email before duplicate lookup.
TC-ADM-USR-12	generates a temporary password	The system successfully generates a temporary password.
TC-ADM-USR-13	hashes the temporary password	The system successfully hashes the temporary password.
TC-ADM-USR-14	requires password change on first login	The system successfully requires password change on first login.
TC-ADM-USR-15	assigns the USER role explicitly	The system successfully assigns the USER role explicitly.
TC-ADM-USR-16	creates an optional address	The system successfully creates an optional address.
TC-ADM-USR-17	creates resident without an address	The system successfully creates resident without an address.
TC-ADM-USR-18	handles password hashing failure	The system handles password hashing failure without producing an uncontrolled failure.
TC-ADM-USR-19	handles user insertion failure	The system handles user insertion failure without producing an uncontrolled failure.
TC-ADM-USR-20	handles address creation failure	The system handles address creation failure without producing an uncontrolled failure.
TC-ADM-USR-21	handles malformed JSON	The system handles malformed JSON without producing an uncontrolled failure.

Test ID	Test scenario	Expected outcome
TC-ADM-USR-22	returns expected success response	The system returns expected success response.
TC-ADM-USR-23	does not log temporary passwords	The system does not log temporary passwords.
TC-ADM-USR-24	propagates address failure from the transaction	The system behaves as specified: propagates address failure from the transaction.
TC-ADM-WASTE-01	creates a valid waste type	The system successfully creates a valid waste type.
TC-ADM-WASTE-02	rejects unauthenticated users	The system rejects unauthenticated users, without allowing the prohibited action.
TC-ADM-WASTE-03	rejects resident users	The system rejects resident users, without allowing the prohibited action.
TC-ADM-WASTE-04	rejects collector users	The system rejects collector users, without allowing the prohibited action.
TC-ADM-WASTE-05	rejects a missing name	The system rejects a missing name, without allowing the prohibited action.
TC-ADM-WASTE-06	rejects a missing price	The system rejects a missing price, without allowing the prohibited action.
TC-ADM-WASTE-07	rejects a zero price	The system rejects a zero price, without allowing the prohibited action.
TC-ADM-WASTE-08	rejects a negative price	The system rejects a negative price, without allowing the prohibited action.
TC-ADM-WASTE-09	rejects a non-numeric price	The system rejects a non-numeric price, without allowing the prohibited action.
TC-ADM-WASTE-10	rejects a whitespace-only name	The system rejects a whitespace-only name, without allowing the prohibited action.

Test ID	Test scenario	Expected outcome
TC-ADM-WASTE-11	rejects an existing waste type	The system rejects an existing waste type, without allowing the prohibited action.
TC-ADM-WASTE-12	defaults to INACTIVE when isActive is omitted	The system defaults to INACTIVE when isActive is omitted.
TC-ADM-WASTE-13	creates an ACTIVE waste type when isActive is true	The system successfully creates an ACTIVE waste type when isActive is true.
TC-ADM-WASTE-14	creates an INACTIVE waste type when isActive is false	The system successfully creates an INACTIVE waste type when isActive is false.
TC-ADM-WASTE-15	saves the trimmed description correctly	The system behaves as specified: saves the trimmed description correctly.
TC-ADM-WASTE-16	uploads an image to Cloudinary	The system behaves as specified: uploads an image to Cloudinary.
TC-ADM-WASTE-17	creates a waste type without an image	The system successfully creates a waste type without an image.
TC-ADM-WASTE-18	handles Cloudinary upload failure	The system handles Cloudinary upload failure without producing an uncontrolled failure.
TC-ADM-WASTE-19	handles database insertion failure	The system handles database insertion failure without producing an uncontrolled failure.
TC-ADM-WASTE-20	handles duplicate lookup failure	The system handles duplicate lookup failure without producing an uncontrolled failure.
TC-ADM-WASTE-21	handles malformed multipart form data	The system handles malformed multipart form data without producing an uncontrolled failure.
TC-ADM-WASTE-22	returns the created waste type in the response	The system returns the created waste type in the response.

Test ID	Test scenario	Expected outcome
TC-ADM-DET-01	retrieves an existing request successfully	The system successfully retrieves an existing request successfully.
TC-ADM-DET-02	rejects unauthenticated users	The system rejects unauthenticated users, without allowing the prohibited action.
TC-ADM-DET-03	rejects non-admin users	The system rejects non-admin users, without allowing the prohibited action.
TC-ADM-DET-04	rejects an empty request ID	The system rejects an empty request ID, without allowing the prohibited action.
TC-ADM-DET-05	rejects a whitespace-only request ID	The system rejects a whitespace-only request ID, without allowing the prohibited action.
TC-ADM-DET-06	returns 404 for a nonexistent request	The system returns 404 for a nonexistent request.
TC-ADM-DET-07	selects the required request relationships	The system successfully selects the required request relationships.
TC-ADM-DET-08	formats the resident	The system successfully formats the resident.
TC-ADM-DET-09	formats the assigned collector	The system successfully formats the assigned collector.
TC-ADM-DET-10	returns null when no collector is assigned	The system returns null when no collector is assigned.
TC-ADM-DET-11	formats the pickup address correctly	The system successfully formats the pickup address correctly.
TC-ADM-DET-12	excludes missing optional address fields from formatting	The system successfully excludes missing optional address fields from formatting.
TC-ADM-DET-13	returns saved address details	The system returns saved address details.
TC-ADM-DET-14	returns waste type details	The system returns waste type details.

Test ID	Test scenario	Expected outcome
TC-ADM-DET-15	returns payment details when available	The system returns payment details when available.
TC-ADM-DET-16	returns null when no payment exists	The system returns null when no payment exists.
TC-ADM-DET-17	returns request activity history	The system returns request activity history.
TC-ADM-DET-18	formats the activity creator	The system successfully formats the activity creator.
TC-ADM-DET-19	handles an activity without a creator	The system handles an activity without a creator without producing an uncontrolled failure.
TC-ADM-DET-20	returns an empty activity history when none exists	The system returns an empty activity history when none exists.
TC-ADM-DET-21	returns request status and lifecycle timestamps	The system returns request status and lifecycle timestamps.
TC-ADM-DET-22	returns 500 when the database query fails	The system returns 500 when the database query fails.
TC-ADM-GET-01	retrieves requests successfully	The system successfully retrieves requests successfully.
TC-ADM-GET-02	returns 401 for unauthenticated users	The system returns 401 for unauthenticated users.
TC-ADM-GET-03	returns 403 for Residents	The system returns 403 for Residents.
TC-ADM-GET-04	applies default pagination	The system successfully applies default pagination.
TC-ADM-GET-05	applies custom pagination	The system successfully applies custom pagination.
TC-ADM-GET-06	caps the limit at 100	The system successfully caps the limit at 100.
TC-ADM-GET-07	defaults invalid nonpositive pagination values	The system defaults invalid nonpositive pagination values.
TC-ADM-GET-08	applies a valid status filter	The system successfully applies a valid status filter.
TC-ADM-GET-09	ignores an invalid status filter	The system behaves as specified: ignores an invalid status filter.

Test ID	Test scenario	Expected outcome
TC-ADM-GET-10	builds case-insensitive search conditions	The system behaves as specified: builds case-insensitive search conditions.
TC-ADM-GET-11	ignores blank search input	The system behaves as specified: ignores blank search input.
TC-ADM-GET-12	applies ascending sorting	The system successfully applies ascending sorting.
TC-ADM-GET-13	defaults invalid sorting to descending	The system defaults invalid sorting to descending.
TC-ADM-GET-14	retrieves requests without a Resident restriction	The system successfully retrieves requests without a Resident restriction.
TC-ADM-GET-15	formats related request information	The system successfully formats related request information.
TC-ADM-GET-16	returns null for an unassigned collector	The system returns null for an unassigned collector.
TC-ADM-GET-17	calculates summary statistics	The system successfully calculates summary statistics.
TC-ADM-GET-18	calculates pagination metadata	The system successfully calculates pagination metadata.
TC-ADM-GET-19	returns an empty result when no requests exist	The system returns an empty result when no requests exist.
TC-ADM-GET-20	returns 500 when a database query fails	The system returns 500 when a database query fails.
TC-ADM-GET-21	defaults malformed page input	The system defaults malformed page input.
TC-ADM-GET-22	defaults malformed limit input	The system defaults malformed limit input.
TC-ADM-GETUSR-01	returns residents successfully	The system returns residents successfully.
TC-ADM-GETUSR-02	rejects unauthenticated requests	The system rejects unauthenticated requests, without allowing the prohibited action.
TC-ADM-GETUSR-03	rejects resident users	The system rejects resident users, without allowing the prohibited action.

Test ID	Test scenario	Expected outcome
TC-ADM-GETUSR-04	rejects collector users	The system rejects collector users, without allowing the prohibited action.
TC-ADM-GETUSR-05	retrieves only USER accounts	The system successfully retrieves only USER accounts.
TC-ADM-GETUSR-06	filters by first name	The system successfully filters by first name.
TC-ADM-GETUSR-07	filters by last name	The system successfully filters by last name.
TC-ADM-GETUSR-08	matches both parts of a full name	The system successfully matches both parts of a full name.
TC-ADM-GETUSR-09	trims whitespace from name filter	The system successfully trims whitespace from name filter.
TC-ADM-GETUSR-10	handles extra spaces in full names	The system handles extra spaces in full names without producing an uncontrolled failure.
TC-ADM-GETUSR-11	ignores empty name filters	The system behaves as specified: ignores empty name filters.
TC-ADM-GETUSR-12	filters by email	The system successfully filters by email.
TC-ADM-GETUSR-13	trims email filter whitespace	The system successfully trims email filter whitespace.
TC-ADM-GETUSR-14	uses case-insensitive email matching	The system successfully uses case-insensitive email matching.
TC-ADM-GETUSR-15	filters ACTIVE residents	The system successfully filters ACTIVE residents.
TC-ADM-GETUSR-16	filters PENDING residents	The system successfully filters PENDING residents.
TC-ADM-GETUSR-17	rejects unsupported status values	The system rejects unsupported status values, without allowing the prohibited action.

Test ID	Test scenario	Expected outcome
TC-ADM-GETUSR-18	accepts lowercase status values	The system behaves as specified: accepts lowercase status values.
TC-ADM-GETUSR-19	combines name, email and status filters	The system behaves as specified: combines name, email and status filters.
TC-ADM-GETUSR-20	sorts residents by newest first	The system successfully sorts residents by newest first.
TC-ADM-GETUSR-21	combines first and last names	The system behaves as specified: combines first and last names.
TC-ADM-GETUSR-22	maps mustChangePassword =false to ACTIVE	The system behaves as specified: maps mustChangePassword =false to ACTIVE.
TC-ADM-GETUSR-23	maps mustChangePassword =true to PENDING	The system behaves as specified: maps mustChangePassword =true to PENDING.
TC-ADM-GETUSR-24	returns only the expected response fields	The system returns only the expected response fields.
TC-ADM-GETUSR-25	excludes sensitive database fields	The system successfully excludes sensitive database fields.
TC-ADM-GETUSR-26	returns an empty array when no users match	The system returns an empty array when no users match.
TC-ADM-GETUSR-27	handles database failures	The system handles database failures without producing an uncontrolled failure.
TC-ADM-GETUSR-28	handles authentication service failures	The system handles authentication service failures without producing an uncontrolled failure.
TC-RES-ADDR-17	successfully updates an owned address	The system behaves as specified: successfully updates an owned address.
TC-RES-ADDR-18	rejects unauthenticated address updates	The system rejects unauthenticated address updates, without allowing the prohibited action.

Test ID	Test scenario	Expected outcome
TC-RES-ADDR-19	rejects address updates by an Administrator	The system rejects address updates by an Administrator, without allowing the prohibited action.
TC-RES-ADDR-20	rejects updates to addresses not owned by the Resident	The system rejects updates to addresses not owned by the Resident, without allowing the prohibited action.
TC-RES-ADDR-21	rejects updates with missing required fields	The system rejects updates with missing required fields, without allowing the prohibited action.
TC-RES-ADDR-22	rejects updates with invalid field types	The system rejects updates with invalid field types, without allowing the prohibited action.
TC-RES-ADDR-23	rejects updates with a short street address	The system rejects updates with a short street address, without allowing the prohibited action.
TC-RES-ADDR-24	rejects updates with a short city name	The system rejects updates with a short city name, without allowing the prohibited action.
TC-RES-ADDR-25	handles a database update failure	The system handles a database update failure without producing an uncontrolled failure.
TC-RES-ADDR-26	rejects a null JSON update body with HTTP 400	The system rejects a null JSON update body with HTTP 400, without allowing the prohibited action.
TC-RES-ADDR-27	successfully deletes an unlinked owned address	The system behaves as specified: successfully deletes an unlinked owned address.
TC-RES-ADDR-28	rejects unauthenticated address deletion	The system rejects unauthenticated address deletion, without allowing the prohibited action.

Test ID	Test scenario	Expected outcome
TC-RES-ADDR-29	rejects address deletion by an Administrator	The system rejects address deletion by an Administrator, without allowing the prohibited action.
TC-RES-ADDR-30	rejects deletion of an address not owned by the Resident	The system rejects deletion of an address not owned by the Resident, without allowing the prohibited action.
TC-RES-ADDR-31	prevents deletion of an address linked to waste requests	The system prevents deletion of an address linked to waste requests, without allowing the prohibited action.
TC-RES-ADDR-32	handles a database deletion failure	The system handles a database deletion failure without producing an uncontrolled failure.
TC-RES-ADDR-01	retrieves addresses belonging to the authenticated Resident	The system successfully retrieves addresses belonging to the authenticated Resident.
TC-RES-ADDR-02	returns an empty array when the Resident has no addresses	The system returns an empty array when the Resident has no addresses.
TC-RES-ADDR-03	rejects unauthenticated address retrieval	The system rejects unauthenticated address retrieval, without allowing the prohibited action.
TC-RES-ADDR-04	rejects address retrieval by an Administrator	The system rejects address retrieval by an Administrator, without allowing the prohibited action.
TC-RES-ADDR-05	returns HTTP 500 when address retrieval fails	The system returns HTTP 500 when address retrieval fails.
TC-RES-ADDR-06	creates a valid Resident address	The system successfully creates a valid Resident address.
TC-RES-ADDR-07	rejects unauthenticated address creation	The system rejects unauthenticated address creation, without allowing the prohibited action.

Test ID	Test scenario	Expected outcome
TC-RES-ADDR-08	rejects address creation by an Administrator	The system rejects address creation by an Administrator, without allowing the prohibited action.
TC-RES-ADDR-09	rejects missing required address fields	The system rejects missing required address fields, without allowing the prohibited action.
TC-RES-ADDR-10	rejects invalid address field types	The system rejects invalid address field types, without allowing the prohibited action.
TC-RES-ADDR-11	rejects street addresses shorter than five characters	The system rejects street addresses shorter than five characters, without allowing the prohibited action.
TC-RES-ADDR-12	rejects a city or state shorter than two characters	The system rejects a city or state shorter than two characters, without allowing the prohibited action.
TC-RES-ADDR-13	stores null when the postal code is omitted	The system successfully stores null when the postal code is omitted.
TC-RES-ADDR-14	defaults the country to Nigeria when omitted	The system defaults the country to Nigeria when omitted.
TC-RES-ADDR-15	returns HTTP 500 when address creation fails	The system returns HTTP 500 when address creation fails.
TC-RES-ADDR-16	rejects a null JSON body with HTTP 400	The system rejects a null JSON body with HTTP 400, without allowing the prohibited action.
TC-RES-CANCEL-01	cancels a PENDING request successfully	The system successfully cancels a PENDING request successfully.
TC-RES-CANCEL-02	cancels an APPROVED request successfully	The system successfully cancels an APPROVED request successfully.

Test ID	Test scenario	Expected outcome
TC-RES-CANCEL-03	rejects unauthenticated cancellation requests	The system rejects unauthenticated cancellation requests, without allowing the prohibited action.
TC-RES-CANCEL-04	rejects Administrator cancellation requests	The system rejects Administrator cancellation requests, without allowing the prohibited action.
TC-RES-CANCEL-05	rejects cancellation when the request ID is missing	The system rejects cancellation when the request ID is missing, without allowing the prohibited action.
TC-RES-CANCEL-06	rejects cancellation reasons exceeding 500 characters	The system rejects cancellation reasons exceeding 500 characters, without allowing the prohibited action.
TC-RES-CANCEL-07	rejects cancellation of a request not owned by the Resident	The system rejects cancellation of a request not owned by the Resident, without allowing the prohibited action.
TC-RES-CANCEL-08	rejects cancellation of an already cancelled request	The system rejects cancellation of an already cancelled request, without allowing the prohibited action.
TC-RES-CANCEL-09	rejects cancellation of a COMPLETED request	The system rejects cancellation of a COMPLETED request, without allowing the prohibited action.
TC-RES-CANCEL-10	rejects cancellation of a REJECTED request	The system rejects cancellation of a REJECTED request, without allowing the prohibited action.
TC-RES-CANCEL-11	rejects cancellation of an ASSIGNED request	The system rejects cancellation of an ASSIGNED request, without allowing the prohibited action.

Test ID	Test scenario	Expected outcome
TC-RES-CANCEL-12	rejects cancellation of an IN_PROGRESS request	The system rejects cancellation of an IN_PROGRESS request, without allowing the prohibited action.
TC-RES-CANCEL-13	cancels a request without providing a reason	The system successfully cancels a request without providing a reason.
TC-RES-CANCEL-14	returns HTTP 500 when the cancellation transaction fails	The system returns HTTP 500 when the cancellation transaction fails.
TC-RES-04A	rejects quantity zero	The system rejects quantity zero, without allowing the prohibited action.
TC-RES-04B	rejects negative quantity	The system rejects negative quantity, without allowing the prohibited action.
TC-RES-04C	rejects decimal quantity	The system rejects decimal quantity, without allowing the prohibited action.
TC-RES-09A	rejects a past pickup date	The system rejects a past pickup date, without allowing the prohibited action.
TC-RES-12B	rejects descriptions over 500 characters	The system rejects descriptions over 500 characters, without allowing the prohibited action.
TC-RES-AUTH-01	rejects unauthenticated users	The system rejects unauthenticated users, without allowing the prohibited action.
TC-RES-AUTH-02	rejects Admin users creating resident requests	The system rejects Admin users creating resident requests, without allowing the prohibited action.
TC-RES-03	creates a valid waste request successfully	The system successfully creates a valid waste request successfully.
TC-RES-05A	rejects a nonexistent waste type	The system rejects a nonexistent waste type, without allowing the prohibited action.

Test ID	Test scenario	Expected outcome
TC-RES-05B	rejects an inactive waste type	The system rejects an inactive waste type, without allowing the prohibited action.
TC-RES-06A	rejects a nonexistent address	The system rejects a nonexistent address, without allowing the prohibited action.
TC-RES-06B	rejects an address belonging to another resident	The system rejects an address belonging to another resident, without allowing the prohibited action.
TC-RES-DASH-01	returns dashboard statistics successfully	The system returns dashboard statistics successfully.
TC-RES-DASH-02	returns 401 when unauthenticated	The system returns 401 when unauthenticated.
TC-RES-DASH-03	returns 401 when an Admin accesses the dashboard	The system returns 401 when an Admin accesses the dashboard.
TC-RES-DASH-04	counts only the implemented active statuses	The system successfully counts only the implemented active statuses.
TC-RES-DASH-05	counts only completed requests	The system successfully counts only completed requests.
TC-RES-DASH-06	scopes all dashboard queries to the authenticated Resident	The system behaves as specified: scopes all dashboard queries to the authenticated Resident.
TC-RES-DASH-07	returns zero statistics for an empty dashboard	The system returns zero statistics for an empty dashboard.
TC-RES-DASH-08	sets hasAddress to true when addresses exist	The system behaves as specified: sets hasAddress to true when addresses exist.
TC-RES-DASH-09	requests the five most recent records	The system behaves as specified: requests the five most recent records.
TC-RES-DASH-10	formats recent request details correctly	The system successfully formats recent request details correctly.

Test ID	Test scenario	Expected outcome
TC-RES-DASH-11	returns 500 when a database query fails	The system returns 500 when a database query fails.
TC-RES-DETAIL-01	retrieves an existing Resident request	The system successfully retrieves an existing Resident request.
TC-RES-DETAIL-02	returns 401 when unauthenticated	The system returns 401 when unauthenticated.
TC-RES-DETAIL-03	returns 403 for an Admin	The system returns 403 for an Admin.
TC-RES-DETAIL-04	returns 400 when request ID is missing	The system returns 400 when request ID is missing.
TC-RES-DETAIL-05	returns 404 when request does not exist	The system returns 404 when request does not exist.
TC-RES-DETAIL-06	prevents retrieving another Resident	The system prevents retrieving another Resident, without allowing the prohibited action.
TC-RES-DETAIL-07	formats the pickup address correctly	The system successfully formats the pickup address correctly.
TC-RES-DETAIL-08	includes the assigned collector	The system successfully includes the assigned collector.
TC-RES-DETAIL-09	handles requests without collector or payment	The system handles requests without collector or payment without producing an uncontrolled failure.
TC-RES-DETAIL-10	includes activity creator full names	The system successfully includes activity creator full names.
TC-RES-DETAIL-11	returns 500 when database retrieval fails	The system returns 500 when database retrieval fails.
TC-RES-GET-01	returns 401 when unauthenticated	The system returns 401 when unauthenticated.
TC-RES-GET-02	returns 403 for an Admin	The system returns 403 for an Admin.
TC-RES-GET-03	retrieves Resident requests successfully	The system successfully retrieves Resident requests successfully.

Test ID	Test scenario	Expected outcome
TC-RES-GET-04	returns an empty list when Resident has no requests	The system returns an empty list when Resident has no requests.
TC-RES-GET-05	applies default pagination	The system successfully applies default pagination.
TC-RES-GET-06	applies custom pagination	The system successfully applies custom pagination.
TC-RES-GET-07	caps the page limit at 50	The system successfully caps the page limit at 50.
TC-RES-GET-08	uses defaults for invalid pagination	The system successfully uses defaults for invalid pagination.
TC-RES-GET-09	filters WAITING requests	The system successfully filters WAITING requests.
TC-RES-GET-10	filters UPCOMING requests	The system successfully filters UPCOMING requests.
TC-RES-GET-11	filters HISTORY requests	The system successfully filters HISTORY requests.
TC-RES-GET-12	exact status overrides view filter	The system behaves as specified: exact status overrides view filter.
TC-RES-GET-13	applies search filters to Resident requests	The system successfully applies search filters to Resident requests.
TC-RES-GET-14	sorts requests in ascending order	The system successfully sorts requests in ascending order.
TC-RES-GET-14B	defaults to descending order for invalid input	The system defaults to descending order for invalid input.
TC-RES-GET-15	restricts requests to the authenticated Resident	The system behaves as specified: restricts requests to the authenticated Resident.
TC-RES-GET-16	formats pickup address and collector name	The system successfully formats pickup address and collector name.
TC-RES-GET-17	returns 500 when database retrieval fails	The system returns 500 when database retrieval fails.

Test ID	Test scenario	Expected outcome
TC-RES-GET-18	defaults malformed pagination parameters	The system defaults malformed pagination parameters.
TC-RES-UPDATE-01	updates general request details	The system successfully updates general request details.
TC-RES-UPDATE-02	reschedules the pickup date	The system behaves as specified: reschedules the pickup date.
TC-RES-UPDATE-03	records both activities when details and date change	The system successfully records both activities when details and date change.
TC-RES-UPDATE-04	returns no changes when submitted details are unchanged	The system returns no changes when submitted details are unchanged.
TC-RES-UPDATE-05	returns 401 when unauthenticated	The system returns 401 when unauthenticated.
TC-RES-UPDATE-06	returns 403 for an Admin	The system returns 403 for an Admin.
TC-RES-UPDATE-07	returns 400 for missing request ID	The system returns 400 for missing request ID.
TC-RES-UPDATE-08	returns 400 for malformed JSON	The system returns 400 for malformed JSON.
TC-RES-UPDATE-09	returns 404 for a missing or unowned request	The system returns 404 for a missing or unowned request.
TC-RES-UPDATE-10	prevents editing non-pending requests	The system prevents editing non-pending requests, without allowing the prohibited action.
TC-RES-UPDATE-11	rejects invalid quantities	The system rejects invalid quantities, without allowing the prohibited action.
TC-RES-UPDATE-12	rejects an invalid quantity unit	The system rejects an invalid quantity unit, without allowing the prohibited action.
TC-RES-UPDATE-13	rejects an invalid pickup date	The system rejects an invalid pickup date, without allowing the prohibited action.
TC-RES-UPDATE-14	rejects a past pickup date	The system rejects a past pickup date, without allowing the prohibited action.

Test ID	Test scenario	Expected outcome
TC-RES-UPDATE-15	rejects descriptions longer than 500 characters	The system rejects descriptions longer than 500 characters, without allowing the prohibited action.
TC-RES-UPDATE-16	returns 404 for an unknown waste type	The system returns 404 for an unknown waste type.
TC-RES-UPDATE-17	rejects an inactive waste type	The system rejects an inactive waste type, without allowing the prohibited action.
TC-RES-UPDATE-18	prevents using another Resident	The system prevents using another Resident, without allowing the prohibited action.
TC-RES-UPDATE-19	updates the pickup address snapshot	The system successfully updates the pickup address snapshot.
TC-RES-UPDATE-20	returns 500 for unexpected database failure	The system returns 500 for unexpected database failure.
TC-RES-UPDATE-21	handles Prisma foreign key errors	The system handles Prisma foreign key errors without producing an uncontrolled failure.
TC-RES-UPDATE-22	rejects a null JSON request body	The system rejects a null JSON request body, without allowing the prohibited action.

Integration test cases

Test ID	Test scenario	Expected outcome
TC-INT-ADDR-01	creates and persists a Resident address	The system successfully creates and persists a Resident address.
TC-INT-ADDR-02	retrieves only the authenticated Resident	The system successfully retrieves only the authenticated Resident.
TC-INT-ADDR-03	updates an address and persists the changes	The system successfully updates an address and persists the changes.

Test ID	Test scenario	Expected outcome
TC-INT-ADDR-04	prevents updating another Resident	The system prevents updating another Resident, without allowing the prohibited action.
TC-INT-ADDR-05	deletes an unlinked address from PostgreSQL	The system behaves as specified: deletes an unlinked address from PostgreSQL.
TC-INT-ADDR-06	prevents deleting an address linked to a waste request	The system prevents deleting an address linked to a waste request, without allowing the prohibited action.
TC-INT-ADDR-07	prevents deleting another Resident	The system prevents deleting another Resident, without allowing the prohibited action.
TC-INT-ADM-APP-01	approves a pending request in PostgreSQL	The system successfully approves a pending request in PostgreSQL.
TC-INT-ADM-APP-02	persists the approval timestamp	The system successfully persists the approval timestamp.
TC-INT-ADM-APP-03	creates an approval activity linked to the Admin	The system successfully creates an approval activity linked to the Admin.
TC-INT-ADM-APP-04	prevents approving an already approved request	The system prevents approving an already approved request, without allowing the prohibited action.
TC-INT-ADM-APP-05	rejects approval of a cancelled request	The system rejects approval of a cancelled request, without allowing the prohibited action.
TC-INT-ADM-APP-06	rejects approval of a rejected request	The system rejects approval of a rejected request, without allowing the prohibited action.
TC-INT-ADM-APP-07	returns 404 for a nonexistent request	The system returns 404 for a nonexistent request.

Test ID	Test scenario	Expected outcome
TC-INT-ADM-APP-08	rolls back approval when activity creation fails	The database transaction rolls back approval when activity creation fails; no partial change is retained.
TC-INT-ADM-APP-09	concurrent approvals must not create duplicate activities	The system behaves as specified: concurrent approvals must not create duplicate activities.
TC-INT-ADM-ASG-01	persists successful collector assignment	The system successfully persists successful collector assignment.
TC-INT-ADM-ASG-02	persists assignment timestamp	The system successfully persists assignment timestamp.
TC-INT-ADM-ASG-03	records collector assignment activity	The system successfully records collector assignment activity.
TC-INT-ADM-ASG-04	prevents sequential duplicate assignment	The system prevents sequential duplicate assignment, without allowing the prohibited action.
TC-INT-ADM-ASG-05	rejects assignment of a pending request	The system rejects assignment of a pending request, without allowing the prohibited action.
TC-INT-ADM-ASG-06	rejects an inactive collector	The system rejects an inactive collector, without allowing the prohibited action.
TC-INT-ADM-ASG-07	returns 404 for a nonexistent request	The system returns 404 for a nonexistent request.
TC-INT-ADM-ASG-08	concurrent assignments must not create duplicate activities	The system behaves as specified: concurrent assignments must not create duplicate activities.
TC-INT-ADM-COL-01	persists a collector account with the correct role	The system successfully persists a collector account with the correct role.
TC-INT-ADM-COL-02	stores a hashed password instead of the temporary password	The system successfully stores a hashed password instead of the temporary password.

Test ID	Test scenario	Expected outcome
TC-INT-ADM-COL-03	requires a password change on first login	The system successfully requires a password change on first login.
TC-INT-ADM-COL-04	creates a linked collector profile with the correct zone	The system successfully creates a linked collector profile with the correct zone.
TC-INT-ADM-COL-05	uses the default active status for a new collector profile	The system successfully uses the default active status for a new collector profile.
TC-INT-ADM-COL-06	rejects duplicate email addresses without creating another profile	The system rejects duplicate email addresses without creating another profile, without allowing the prohibited action.
TC-INT-ADM-COL-07	normalises email before saving it	The system successfully normalises email before saving it.
TC-INT-ADM-COL-08	rejects an invalid email without saving an account	The system rejects an invalid email without saving an account, without allowing the prohibited action.
TC-INT-ADM-COL-09	rejects whitespace-only zone without saving an account	The system rejects whitespace-only zone without saving an account, without allowing the prohibited action.
TC-INT-ADM-COL-10	rejects unauthenticated requests without saving an account	The system rejects unauthenticated requests without saving an account, without allowing the prohibited action.
TC-INT-ADM-COL-11	rejects resident users without saving an account	The system rejects resident users without saving an account, without allowing the prohibited action.
TC-INT-ADM-COL-12	rejects malformed JSON	The system rejects malformed JSON, without allowing the prohibited action.

Test ID	Test scenario	Expected outcome
TC-INT-ADM-COL-13	rolls back account creation when collector profile insertion fails	The database transaction rolls back account creation when collector profile insertion fails; no partial change is retained.
TC-INT-ADM-USR-01	creates and persists a resident	The system successfully creates and persists a resident.
TC-INT-ADM-USR-02	stores a hashed password	The system successfully stores a hashed password.
TC-INT-ADM-USR-03	requires password change	The system successfully requires password change.
TC-INT-ADM-USR-04	persists the optional address	The system successfully persists the optional address.
TC-INT-ADM-USR-05	creates a resident without an address	The system successfully creates a resident without an address.
TC-INT-ADM-USR-06	rejects duplicate email	The system rejects duplicate email, without allowing the prohibited action.
TC-INT-ADM-USR-07	normalises email before persistence	The system successfully normalises email before persistence.
TC-INT-ADM-USR-08	rejects invalid email without insertion	The system rejects invalid email without insertion, without allowing the prohibited action.
TC-INT-ADM-USR-09	rejects whitespace-only names	The system rejects whitespace-only names, without allowing the prohibited action.
TC-INT-ADM-USR-10	rejects unauthenticated creation	The system rejects unauthenticated creation, without allowing the prohibited action.
TC-INT-ADM-USR-11	rejects non-admin creation	The system rejects non-admin creation, without allowing the prohibited action.

Test ID	Test scenario	Expected outcome
TC-INT-ADM-USR-12	rolls back user creation when address insertion fails	The database transaction rolls back user creation when address insertion fails; no partial change is retained.
TC-INT-ADM-WASTE-01	persists a valid active waste type	The system successfully persists a valid active waste type.
TC-INT-ADM-WASTE-02	persists an inactive waste type	The system successfully persists an inactive waste type.
TC-INT-ADM-WASTE-03	trims name and description before saving	The system successfully trims name and description before saving.
TC-INT-ADM-WASTE-04	rejects a duplicate waste type	The system rejects a duplicate waste type, without allowing the prohibited action.
TC-INT-ADM-WASTE-05	rejects negative prices without insertion	The system rejects negative prices without insertion, without allowing the prohibited action.
TC-INT-ADM-WASTE-06	rejects whitespace-only names without insertion	The system rejects whitespace-only names without insertion, without allowing the prohibited action.
TC-INT-ADM-WASTE-07	rejects unauthorised creation	The system rejects unauthorised creation, without allowing the prohibited action.
TC-INT-ADM-WASTE-08	retrieves a newly created waste type from PostgreSQL	The system successfully retrieves a newly created waste type from PostgreSQL.
TC-INT-ADM-DET-01	retrieves request, resident and waste type	The system successfully retrieves request, resident and waste type.
TC-INT-ADM-DET-02	returns pickup snapshot and saved address separately	The system returns pickup snapshot and saved address separately.
TC-INT-ADM-DET-03	retrieves assigned collector and profile	The system successfully retrieves assigned collector and profile.

Test ID	Test scenario	Expected outcome
TC-INT-ADM-DET-04	retrieves persisted payment details	The system successfully retrieves persisted payment details.
TC-INT-ADM-DET-05	returns activity history newest first	The system returns activity history newest first.
TC-INT-ADM-DET-06	handles missing collector and payment	The system handles missing collector and payment without producing an uncontrolled failure.
TC-INT-ADM-DET-07	returns 404 for nonexistent request	The system returns 404 for nonexistent request.
TC-INT-ADM-DET-08	does not expose password fields	The system does not expose password fields.
TC-INT-ADM-GET-01	retrieves requests from multiple Residents	The system successfully retrieves requests from multiple Residents.
TC-INT-ADM-GET-02	filters requests by status	The system successfully filters requests by status.
TC-INT-ADM-GET-03	searches by request reference	The system successfully searches by request reference.
TC-INT-ADM-GET-04	searches by Resident name	The system successfully searches by Resident name.
TC-INT-ADM-GET-05	applies pagination and descending sorting	The system successfully applies pagination and descending sorting.
TC-INT-ADM-GET-06	returns formatted related records	The system returns formatted related records.
TC-INT-ADM-GET-07	returns accurate global statistics	The system returns accurate global statistics.
TC-INT-ADM-GETUSR-01	retrieves resident accounts	The system successfully retrieves resident accounts.
TC-INT-ADM-GETUSR-02	excludes admins and collectors	The system successfully excludes admins and collectors.
TC-INT-ADM-GETUSR-03	searches by first name	The system successfully searches by first name.
TC-INT-ADM-GETUSR-04	searches by last name	The system successfully searches by last name.

Test ID	Test scenario	Expected outcome
TC-INT-ADM-GETUSR-05	matches both full-name components	The system successfully matches both full-name components.
TC-INT-ADM-GETUSR-06	handles extra spaces in full names	The system handles extra spaces in full names without producing an uncontrolled failure.
TC-INT-ADM-GETUSR-07	searches email case-insensitively	The system successfully searches email case-insensitively.
TC-INT-ADM-GETUSR-08	filters ACTIVE residents	The system successfully filters ACTIVE residents.
TC-INT-ADM-GETUSR-09	filters PENDING residents	The system successfully filters PENDING residents.
TC-INT-ADM-GETUSR-10	combines multiple filters	The system behaves as specified: combines multiple filters.
TC-INT-ADM-GETUSR-11	rejects invalid status filters	The system rejects invalid status filters, without allowing the prohibited action.
TC-INT-ADM-GETUSR-12	enforces access restrictions	The system successfully enforces access restrictions.
TC-INT-ADM-GETUSR-13	excludes sensitive response fields	The system successfully excludes sensitive response fields.
TC-INT-ADM-GETUSR-14	returns empty results for unmatched searches	The system returns empty results for unmatched searches.
TC-INT-CANCEL-01	persists cancellation of a PENDING request	The system successfully persists cancellation of a PENDING request.
TC-INT-CANCEL-02	persists cancellation of an APPROVED request	The system successfully persists cancellation of an APPROVED request.
TC-INT-CANCEL-03	persists a cancellation activity in PostgreSQL	The system successfully persists a cancellation activity in PostgreSQL.

Test ID	Test scenario	Expected outcome
TC-INT-CANCEL-04	prevents cancellation of another Resident	The system prevents cancellation of another Resident, without allowing the prohibited action.
TC-INT-CANCEL-05	rolls back cancellation when activity creation fails	The database transaction rolls back cancellation when activity creation fails; no partial change is retained.
TC-INT-01	persists a valid Resident waste request	The system successfully persists a valid Resident waste request.
TC-INT-03	rejects another Resident	The system rejects another Resident, without allowing the prohibited action.
TC-INT-04	rolls back request when activity creation fails	The database transaction rolls back request when activity creation fails; no partial change is retained.
TC-INT-DASH-01	retrieves accurate dashboard statistics from PostgreSQL	The system successfully retrieves accurate dashboard statistics from PostgreSQL.
TC-INT-DASH-02	excludes another Resident	The system successfully excludes another Resident.
TC-INT-DASH-03	returns zero statistics for a Resident without records	The system returns zero statistics for a Resident without records.
TC-INT-DASH-04	updates hasAddress based on saved addresses	The system successfully updates hasAddress based on saved addresses.
TC-INT-DASH-05	returns only five most recent requests in descending order	The system returns only five most recent requests in descending order.
TC-INT-DASH-06	classifies request statuses according to the current implementation	The system behaves as specified: classifies request statuses according to the current implementation.

Test ID	Test scenario	Expected outcome
TC-INT-GET-01	retrieves an existing Resident waste request	The system successfully retrieves an existing Resident waste request.
TC-INT-GET-02	prevents a Resident from retrieving another Resident	The system prevents a Resident from retrieving another Resident, without allowing the prohibited action.
TC-INT-GET-03	filters Resident requests by collection view	The system successfully filters Resident requests by collection view.
TC-INT-GET-04	applies pagination and ascending sorting to Resident requests	The system successfully applies pagination and ascending sorting to Resident requests.
TC-INT-GET-05	searches Resident waste requests using PostgreSQL	The system successfully searches Resident waste requests using PostgreSQL.
TC-INT-DETAIL-01	retrieves request details from PostgreSQL	The system successfully retrieves request details from PostgreSQL.
TC-INT-DETAIL-02	prevents another Resident from retrieving request details	The system prevents another Resident from retrieving request details, without allowing the prohibited action.
TC-INT-DETAIL-03	handles absent collector and payment relationships	The system handles absent collector and payment relationships without producing an uncontrolled failure.
TC-INT-UPDATE-01	persists updated details and pickup address	The system successfully persists updated details and pickup address.
TC-INT-UPDATE-02	persists pickup rescheduling activity	The system successfully persists pickup rescheduling activity.
TC-INT-UPDATE-03	persists both activities when details and date change	The system successfully persists both activities when details and date change.

Test ID	Test scenario	Expected outcome
TC-INT-UPDATE-04	prevents editing an APPROVED request	The system prevents editing an APPROVED request, without allowing the prohibited action.
TC-INT-UPDATE-05	prevents editing another Resident	The system prevents editing another Resident, without allowing the prohibited action.
TC-INT-UPDATE-06	rolls back update when activity creation fails	The database transaction rolls back update when activity creation fails; no partial change is retained.

End-to-End test cases

Test ID	Test scenario	Expected outcome
TC-E2E-ADMIN-01	administrator logs in successfully	The system behaves as specified: administrator logs in successfully.
TC-E2E-ADMIN-02	displays request management	The system successfully displays request management.
TC-E2E-ADMIN-03	finds a pending resident request	The system successfully finds a pending resident request.
TC-E2E-ADMIN-04	views resident request details	The system behaves as specified: views resident request details.
TC-E2E-ADMIN-05	approves a pending pickup request	The system successfully approves a pending pickup request.
TC-E2E-ADMIN-06	assigns an active collector	The system successfully assigns an active collector.
TC-E2E-LOGIN-01	displays the login form	The system successfully displays the login form.
TC-E2E-AUTH-01	resident logs in successfully	The system behaves as specified: resident logs in successfully.
TC-E2E-AUTH-02	rejects incorrect password	The system rejects incorrect password, without allowing the prohibited action.

Test ID	Test scenario	Expected outcome
TC-E2E-AUTH-03	rejects unregistered email	The system rejects unregistered email, without allowing the prohibited action.
TC-E2E-AUTH-04	validates empty login fields	The application validates empty login fields and displays the relevant validation response.
TC-E2E-AUTH-05	prevents resident from accessing admin dashboard	The system prevents resident from accessing admin dashboard, without allowing the prohibited action.
TC-E2E-AUTH-06	redirects unauthenticated user to login	The application redirects unauthenticated user to login.
TC-E2E-AUTH-07	redirects password-change-required resident	The application redirects password-change-required resident.
TC-E2E-PICKUP-01	prompts resident to add an address	The system successfully prompts resident to add an address.
TC-E2E-PICKUP-02	creates a pickup address through the UI	The system successfully creates a pickup address through the UI.
TC-E2E-PICKUP-03	rejects an invalid quantity	The system rejects an invalid quantity, without allowing the prohibited action.
TC-E2E-PICKUP-04	rejects a past pickup date	The system rejects a past pickup date, without allowing the prohibited action.
TC-E2E-PICKUP-05	submits a valid pickup request	The system successfully submits a valid pickup request.
TC-E2E-PICKUP-06	displays submitted request under Waiting	The system successfully displays submitted request under Waiting.

Performance test scenarios

Test ID	Test scenario	Expected outcome
PERF-01	Baseline load using k6 baseline.js	The configured HTTP requests complete within the script

		thresholds (p95 below 2,000 ms and HTTP failure rate below 1%).
PERF-02	Concurrent load using k6 concurrent-load.js	The application remains within the configured response-time and failure-rate thresholds under concurrent requests.
PERF-03	Increased load using k6 increased-load.js	The application remains within the configured response-time and failure-rate thresholds for the increased workload.
PERF-04	Authenticated requests using k6 authenticated-requests.js	Authenticated requests meet the configured response-time and failure-rate thresholds.

The source tests are located in tests/api/, tests/integration/, tests/e2e/ and tests/performance/.